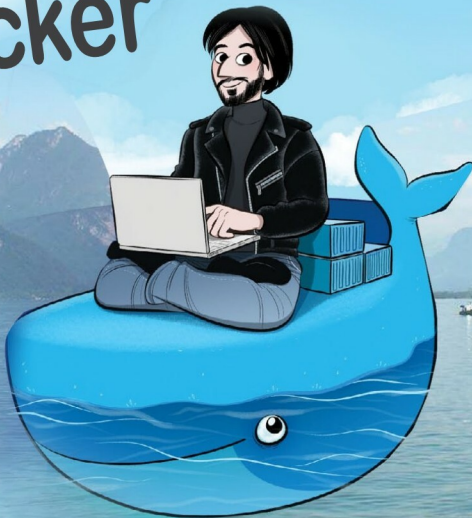


An Introduction to Docker

Learn about Docker, which is an open source project that can automate the deployment of software applications inside software containers. Docker can be provisioned on any infrastructure, be it laptops or the public cloud. This platform can be very useful for sysadmins and developers.



Docker is an open source platform for development and operation teams to transport, build and run distributed applications in an effective manner. It can combine an application along with its dependencies in a virtual container that can run on a Linux server by providing abstraction and automation.

The concept behind using containers is to isolate resources from physical hosts, limit the services and provision them to have a private view of the operating system with their own file system structure, process ID space and network interfaces. A single kernel can have multiple containers but each container can use only a defined amount of I/O, CPU and memory.

This leads to an increase in portability and flexibility for the applications that can run on cloud deployment models such as

private and public clouds, and also on bare metal and so on. Docker consists of:

- Docker Engine: This is a portable packaging tool which has lightweight application runtime.
- Docker Hub: This is a cloud service for application sharing and workflow automation.

Docker uses resource isolation features of the kernel such as kernel namespaces and *cgroups* to allow independent containers to run within a single instance, thereby avoiding the overhead of starting virtual machines.

The Linux kernel's namespace provides isolation to the application's view of the operating environment, which includes the network, process trees, mounted file systems and